

Recognising Torus Knots from the Alexander Polynomial

Justification for the `torus_match` decision procedure

Knoodle project

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Abstract

This note explains why the Alexander polynomial of a torus knot $T(p, q)$ has the closed form $(t^{pq} - 1)(t - 1) / ((t^p - 1)(t^q - 1))$, why that closed form yields a *finite and exhaustive* test for torus-knot-hood at a given degree, and exactly where the program `torus_match.cpp` converts that test into a verdict. The logical asymmetry of the two verdicts is emphasised: a negative answer (“not a torus knot”) is a proof, whereas a positive answer is only strong evidence. We then apply the criterion to the constant-difference petal permutations of length $n = 13$ and show that four of the twelve are provably not torus knots, contradicting the conjecture recorded in the header of `sample_const_diff.cpp`.

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1 Setting and notation

For coprime integers $1 < p < q$, the torus knot $T(p, q)$ is the simple closed curve on the standardly embedded torus in S^3 that wraps p times in the meridional direction and q times in the longitudinal direction. Coprimality is what makes the curve connected; if $\gcd(p, q) = g > 1$ the same recipe

produces a g -component link rather than a knot. Reversing the roles of p and q gives the same knot, so we fix the canonical order $p < q$.

The Alexander polynomial $\Delta_K(t)$ of a knot K is well defined only up to multiplication by a unit $\pm t^k$ of $\mathbb{Z}[t, t^{-1}]$. Throughout we work with the *normalised* representative: the unique associate that is an honest polynomial with non-zero constant term and positive lowest coefficient. Two facts we use as standing sanity conditions are that $\Delta_K(1) = \pm 1$ and that Δ_K is palindromic, i.e. $\Delta_K(t) = \pm t^{\deg \Delta_K} \Delta_K(1/t)$.

We write $\Phi_d(t)$ for the d -th cyclotomic polynomial, so that

$$t^n - 1 = \prod_{d|n} \Phi_d(t), \quad \deg \Phi_d = \varphi(d). \quad (1)$$

2 The closed form

2.1 Topological input

The single non-elementary ingredient is Milnor's fibration theorem applied to the Brieskorn singularity $f(z_1, z_2) = z_1^p + z_2^q$. The link of this singularity, $f^{-1}(0) \cap S_\varepsilon^3$, is precisely $T(p, q)$. Milnor shows that the complement fibres over S^1 , that the fibre F (the Milnor fibre) has the homotopy type of a wedge of $\mu = (p-1)(q-1)$ circles, and that the characteristic polynomial of the monodromy $h_*: H_1(F) \rightarrow H_1(F)$ is the Alexander polynomial of the link. The eigenvalues of h_* are the products of non-trivial roots of unity

$$\zeta_p^i \zeta_q^j, \quad 1 \leq i \leq p-1, \quad 1 \leq j \leq q-1, \quad (2)$$

where $\zeta_m = e^{2\pi i/m}$. Consequently

$$\Delta_{T(p,q)}(t) = \prod_{i=1}^{p-1} \prod_{j=1}^{q-1} (t - \zeta_p^i \zeta_q^j). \quad (3)$$

Everything below is elementary algebra applied to (3). (An alternative derivation, avoiding singularity theory, computes a Seifert matrix for the standard genus- $\frac{1}{2}(p-1)(q-1)$ fibre surface, or reduces the Burau representation of the braid $(\sigma_1 \cdots \sigma_{p-1})^q$; both give the same answer.)

2.2 Summing the product

Theorem 1. *For coprime $1 < p < q$,*

$$\Delta_{T(p,q)}(t) = \frac{(t^{pq} - 1)(t - 1)}{(t^p - 1)(t^q - 1)},$$

and the right-hand side is a polynomial with integer coefficients of degree $(p-1)(q-1)$.

Proof. Consider the map

$$\alpha: \mathbb{Z}/p \times \mathbb{Z}/q \longrightarrow \mu_{pq}, \quad (i, j) \longmapsto \zeta_p^i \zeta_q^j,$$

where μ_{pq} denotes the group of pq -th roots of unity. Because $\gcd(p, q) = 1$, the Chinese Remainder Theorem makes α a group isomorphism; in particular it is a bijection. Hence as (i, j) ranges over all pq pairs, $\zeta_p^i \zeta_q^j$ ranges over each pq -th root of unity exactly once, giving

$$\prod_{i=0}^{p-1} \prod_{j=0}^{q-1} (t - \zeta_p^i \zeta_q^j) = \prod_{\omega \in \mu_{pq}} (t - \omega) = t^{pq} - 1. \quad (4)$$

Now split the index set $\{0, \dots, p-1\} \times \{0, \dots, q-1\}$ into four blocks according to whether $i = 0$ and whether $j = 0$:

$$\begin{aligned} i = 0, j = 0 : & \quad t - 1, \\ i = 0, j \neq 0 : & \quad \prod_{j=1}^{q-1} (t - \zeta_q^j) = \frac{t^q - 1}{t - 1}, \\ i \neq 0, j = 0 : & \quad \prod_{i=1}^{p-1} (t - \zeta_p^i) = \frac{t^p - 1}{t - 1}, \\ i \neq 0, j \neq 0 : & \quad P(t) := \prod_{i=1}^{p-1} \prod_{j=1}^{q-1} (t - \zeta_p^i \zeta_q^j). \end{aligned}$$

Multiplying the four blocks and equating with (4),

$$t^{pq} - 1 = (t - 1) \cdot \frac{t^q - 1}{t - 1} \cdot \frac{t^p - 1}{t - 1} \cdot P(t) = \frac{(t^p - 1)(t^q - 1)}{t - 1} \cdot P(t).$$

Solving for P and comparing with (3) gives the stated formula. The product P has $(p-1)(q-1)$ linear factors, so the degree is as claimed; it lies in $\mathbb{Z}[t]$ because it is a quotient of monic integer polynomials that divides exactly. $\square$

Corollary 2 (Cyclotomic form). $\Delta_{T(p,q)}(t) = \prod_{\substack{d|pq \\ d \nmid p, d \nmid q}} \Phi_d(t).$

Proof. Since $\gcd(p, q) = 1$, every divisor of pq factors uniquely as $d_1 d_2$ with $d_1 \mid p$ and $d_2 \mid q$; moreover $\{d : d \mid p\} \cap \{d : d \mid q\} = \{1\}$. Partition the divisors of pq into those dividing p , those dividing q and exceeding 1, and the rest. Using (1) and $\Phi_1(t) = t - 1$,

$$t^{pq} - 1 = \prod_{d|pq} \Phi_d = \underbrace{\left(\prod_{d|p} \Phi_d \right)}_{t^p - 1} \underbrace{\left(\prod_{\substack{d|q \\ d > 1}} \Phi_d \right)}_{(t^q - 1)/(t - 1)} \prod_{\substack{d|pq \\ d \nmid p, d \nmid q}} \Phi_d.$$

Substituting into Theorem 1 cancels $(t^p - 1)(t^q - 1)/(t - 1)$ against the same factor, leaving exactly the asserted product. $\square$

Corollary 2 is the form used as an independent cross-check in the verification script: computing $\Delta_{T(p,q)}$ by exact division and by multiplying cyclotomics are genuinely different code paths, and they were confirmed to agree on all 163 torus knots with $p \leq 13$, $q < 30$.

Corollary 3 (Degree and genus). $\deg \Delta_{T(p,q)} = (p-1)(q-1)$. *Since torus knots are fibred, the Seifert genus satisfies $g(T(p, q)) = \frac{1}{2}(p-1)(q-1)$ and $\deg \Delta = 2g$.*

Proof. Immediate from Theorem 1. Via Corollary 2 it also follows from $\sum_{d|n} \varphi(d) = n$ and inclusion-exclusion: $\sum \varphi(d) = pq - p - q + 1 = (p-1)(q-1)$, the term $+1$ correcting for $d = 1$ being counted in both $\{d \mid p\}$ and $\{d \mid q\}$. $\square$

Lemma 4 (Parity). $\deg \Delta_{T(p,q)}$ is always even.

Proof. p and q are coprime, so they are not both even. If both are odd, $(p-1)(q-1)$ is a product of two even numbers. If exactly one is even, say p , then $q-1$ is even. In either case the product is even. (This also follows from $\deg \Delta = 2g$.) $\square$

3 The decision procedure

3.1 Finiteness of the candidate set

The whole method rests on the following observation: the degree of the input pins the candidates down to a handful.

Theorem 5 (Exhaustive candidate enumeration). *Let K be a knot with normalised Alexander polynomial of degree $D > 0$. If $K \cong T(p, q)$ for some coprime $1 < p < q$, then*

$$(p-1) \mid D, \quad q = \frac{D}{p-1} + 1, \quad 2 \leq p < \sqrt{D} + 1.$$

In particular the number of candidate pairs is at most the number of divisors of D , and is bounded by $\sqrt{D}$.

Proof. By Corollary 3, $D = (p-1)(q-1)$, so $p-1$ divides D and $q-1 = D/(p-1)$, giving the stated q . The canonical order $p < q$ is equivalent to $p-1 < q-1 = D/(p-1)$, i.e. $(p-1)^2 < D$, i.e. $p < \sqrt{D} + 1$. $\square$

This is what makes a negative verdict a genuine proof rather than a failed search: the list of pairs to test is complete, not heuristic.

Proposition 6 (Asymmetry of the verdicts). *Let K have normalised Alexander polynomial Δ_K of degree D , and let $\mathcal{C}(D)$ be the candidate set of Theorem 5.*

1. *If $\Delta_K \neq \Delta_{T(p,q)}$ for every $(p, q) \in \mathcal{C}(D)$, then K is not a torus knot. This conclusion is rigorous.*
2. *If $\Delta_K = \Delta_{T(p,q)}$ for some (p, q) , it does not follow that $K \cong T(p, q)$. The Alexander polynomial does not separate knots.*

Proof. (1) is the contrapositive of Theorem 5 together with the fact that the Alexander polynomial is a knot invariant: were K a torus knot, its parameters would have to lie in $\mathcal{C}(D)$ and its polynomial would have to match. (2) holds because Δ is not a complete invariant — for instance every untwisted Whitehead double has $\Delta = 1$, the same as the unknot. $\square$

Proposition 6 is the reason the summary line of `sample_const_diff` is worded as “PROVABLY NOT torus” for the negative count while the positive count is reported merely as “torus or unknot”.

Remark 7. A caveat that belongs with (1): the rigour is conditional on Δ_K having been computed correctly from a diagram genuinely representing K . The mathematics is airtight; the guarantee is only as strong as the upstream geometry and polynomial code. Section 6 records what was done to test that.

3.2 Normalisation

Because Δ is defined only up to $\pm t^k$, comparison requires a canonical representative on both sides. The procedure divides out the largest power of t , then negates if necessary so that the lowest surviving coefficient is positive. Both the input polynomial and each $\Delta_{T(p,q)}$ are passed through the same normalisation, so equality testing is well defined.

3.3 The algorithm

1. Read the coefficient list of Δ_K , lowest power first; normalise; let D be its degree.
2. If $D = 0$, report *unknot* ($\Delta = 1$).
3. If D is odd, report *not a torus knot* — justified by Lemma 4, and reached without any search.
4. Otherwise, for each p with $2 \leq p \leq p_{\max}$ such that $(p-1) \mid D$, set $q = D/(p-1) + 1$; discard the pair unless $q > p$ and $\gcd(p, q) = 1$; compute $\Delta_{T(p,q)}$ by Theorem 1 and compare.
5. Report every match; if there are none, report *not a torus knot*.

Remark 8 (The bound $p_{\max}$ is not a real restriction). The implementation defaults to $p_{\max} = 64$, which at first sight looks like a truncation that could manufacture false negatives. It cannot, for degrees in any plausible range: the guard $q > p$ already forces $p < \sqrt{D} + 1$ by Theorem 5, so $p_{\max} = 64$ is vacuous for every $D < 63^2 = 3969$. The largest degree arising in the computations below is 30, where the guard caps p at 6.

4 Where the code implements this

The decision procedure lives in `torus_match.cpp`; the caller that turns it into a per-permutation column lives in `sample_const_diff.cpp`.

Location	Role	Correspondence
<i>torus_match.cpp</i> — the decision procedure		
line 99	<code>torusAlexander(p,q)</code>	Evaluates Theorem 1 by exact polynomial division.
line 84	<code>normalize</code>	Canonical representative modulo $\pm t^k$.
line 226	odd-degree exit	Lemma 4. Emits <code>no -- odd degree</code> , exit status 2.
line 238	<code>for (p = 2; p <= maxP; ++p)</code>	Enumerates $\mathcal{C}(D)$.
line 240	<code>if (deg % pm != 0)</code>	Imposes $(p-1) \mid D$.
line 242	<code>if (q <= p) continue;</code>	Canonical order; supplies the $p < \sqrt{D} + 1$ bound.
line 243	<code>if (gcdInt(p,q) != 1)</code>	Rejects links.
line 245	<code>hits.push_back(...)</code>	Records a match.
lines 265–267	<code>if (hits.empty())</code>	The negative verdict. Emits <code>Torus : no -- no T(p,q) has this Alexander polynomial</code> , exit status 2.
<i>sample_const_diff.cpp</i> — the caller		
line 225	<code>torusViaAlexander</code>	Shells out to the pipeline and captures the label.
line 339	<code>else if (torus == "no") ++nonTorus;</code>	Counts negatives.
lines 348–349	summary line	Prints PROBABLY NOT torus .

Note that the two output columns are produced by two independent routes that never cross-check each other: the `knot` column is a lookup in the KLUT table via `knoodleidentify`, whereas the `torus_type` column is the polynomial computation described here. This is deliberate — the polynomial route remains meaningful exactly where the table route fails, namely above the table’s crossing range — but it does mean the two columns can disagree without either being wrong.

5 Worked examples: the constant-difference permutations at $n = 13$

The constant-difference petal permutation of length n with step d , $\gcd(d, n) = 1$, is $\pi_i = ((i - 1)d \bmod n) + 1$. Running the pipeline over all $\varphi(13) = 12$ steps gives the following.

d	KLUT name	Δ degree	verdict	diagram crossings
1	a0.1	0	unknot	0
2	m7.1	6	$T(2, 7)$	7
3	notfound[14]	12	$T(3, 7)$	14
4	unidentified[19]	4	no	19
5	unidentified[26]	10	no	26
6	unidentified[35]	30	$T(6, 7)$	35
7	unidentified[35]	30	$T(6, 7)$	35
8	unidentified[26]	10	no	26
9	unidentified[19]	4	no	19
10	notfound[14]	12	$T(3, 7)$	14
11	p7.1	6	$T(2, 7)$	7
12	a0.1	0	unknot	0

5.1 Why $d = 4$ and $d = 9$ are not torus knots

Both give the normalised polynomial

$$\Delta(t) = t^4 - t^2 + 1 = \Phi_{12}(t), \quad D = 4.$$

By Theorem 5 the divisors of $D = 4$ give $p - 1 \in \{1, 2, 4\}$, i.e. $p \in \{2, 3, 5\}$:

- $p = 2 \Rightarrow q = 5$, coprime, $q > p$: **admissible**, the single candidate;
- $p = 3 \Rightarrow q = 3$, rejected by $q > p$ (indeed $T(3, 3)$ is a three-component link);
- $p = 5 \Rightarrow q = 2$, rejected by $q > p$ (it is $T(2, 5)$ again).

So the candidate set is $\mathcal{C}(4) = \{(2, 5)\}$, and by Corollary 2 $\Delta_{T(2,5)} = \Phi_{10} = t^4 - t^3 + t^2 - t + 1$. Since $\Phi_{12} \neq \Phi_{10}$, no torus knot has this polynomial, and by Proposition 6(1) the knot is provably not a torus knot.

It is worth recording what the polynomial *does* look like. As $\Delta_{3_1}(t) = t^2 - t + 1 = \Phi_6(t)$ and $\Phi_6(t^2) = t^4 - t^2 + 1 = \Phi_{12}(t)$, we have $\Delta(t) = \Delta_{3_1}(t^2)$. A substitution $t \mapsto t^2$ is the signature of a satellite with winding number 2 — the Alexander polynomial of a cable satisfies $\Delta_{C_{a,b}(K)}(t) = \Delta_K(t^a) \Delta_{T(a,b)}(t)$ — so these knots are consistent with being winding-number-2 satellites of the trefoil. That is a plausible structure for a petal construction to produce, and it is not a torus knot.

5.2 Why $d = 5$ and $d = 8$ are not torus knots

Both give

$$\Delta(t) = t^{10} - 14t^9 + 78t^8 - 221t^7 + 390t^6 - 467t^5 + 390t^4 - 221t^3 + 78t^2 - 14t + 1,$$

of degree $D = 10$. The divisors give $p - 1 \in \{1, 2, 5, 10\}$, i.e. $p \in \{2, 3, 6, 11\}$:

- $p = 2 \Rightarrow q = 11$, coprime, $q > p$: **admissible**;
- $p = 3 \Rightarrow q = 6$, rejected since $\gcd(3, 6) = 3 \neq 1$ (a link, not a knot);
- $p = 6 \Rightarrow q = 3$, rejected by $q > p$;
- $p = 11 \Rightarrow q = 2$, rejected by $q > p$.

So $\mathcal{C}(10) = \{(2, 11)\}$ and $\Delta_{T(2,11)} = \Phi_{22} = t^{10} - t^9 + t^8 - \dots - t + 1$, all coefficients ± 1 . The observed polynomial has a coefficient -467 , so it is not Φ_{22} , and the verdict is again a proof.

As consistency checks, $\Delta(1) = 1$ as required, the coefficient sequence is palindromic, and the determinant $|\Delta(-1)| = 1875 = 3 \cdot 5^4$ is odd, as every knot determinant must be.

5.3 Consequence

The header of `sample_const_diff.cpp` describes the program as testing “the conjecture that every constant-difference petal permutation produces a torus knot”. The rows $d \in \{4, 5, 8, 9\}$ at $n = 13$ refute that conjecture, as do $d \in \{4, 7\}$ at $n = 11$ and $d \in \{5, 6, 7, 10, 11, 12\}$ at $n = 17$. Interestingly $n = 5, 7, 15$ produce torus knots throughout. Empirically the torus cases cluster at small d , at $d \approx n/2$, and at the mirrors $d \leftrightarrow n - d$; the failures occupy the bands in between.

6 Verification performed

Because a negative verdict is asserted as a proof, the supporting computation was tested rather than trusted.

1. **Matcher correctness.** $\Delta_{T(p,q)}$ was recomputed by two independent implementations — exact division (Theorem 1) and the cyclotomic product (Corollary 2) — and compared with the program’s verdict for all 163 torus knots with $p \leq 13$, $q < 30$. There were no disagreements, and no two distinct torus knots in that range shared an Alexander polynomial.
2. **Invariance.** The Alexander polynomial is a knot invariant, so it must not depend on which diagram the simplifier happens to produce. Eight independent randomised re-projections of each of several permutations yielded byte-identical coefficient lists in every case, including the degree-30 example.
3. **Agreement with theory.** For the permutations whose verdict is positive, the computed polynomial was compared against the closed form, and the crossing number of the simplified diagram against the known crossing number $c(T(p, q)) = \min\{p(q - 1), q(p - 1)\}$:

case	claim	diagram crossings	$c(T(p, q))$	deg Δ	polynomial
$n = 9, d = 4$	$T(4, 5)$	15	15	12	exact match
$n = 11, d = 5$	$T(5, 6)$	24	24	20	exact match
$n = 13, d = 3$	$T(3, 7)$	14	14	12	exact match
$n = 13, d = 6$	$T(6, 7)$	35	35	30	exact match

Reproducing a 31-coefficient polynomial exactly, with the diagram’s crossing count independently landing on the theoretical value, is strong evidence that the geometry pipeline and the polynomial routine are both faithful.

4. **Axioms.** All 38 polynomials arising for $n \in \{5, 7, 9, 11, 13\}$ satisfy $\Delta(1) = \pm 1$ and are palindromic.

Remaining gaps. Two, stated plainly. First, the Alexander routine was validated *indirectly*, by exact agreement with theory on knots whose answer is known, rather than by an independent reimplementaion of Fox calculus over the PD code; that reimplementaion is the one outstanding belt-and-braces check. Second, all figures above were produced through the local `identify_knot` binary. The mathematics is independent of which program computes Δ , but re-sourcing the polynomial from the library's own `src/KnotInvariants/Alexander.hpp` would remove the last dependence on non-upstream code.

References

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